Phases 1–3: Problem Definition, Training and Spatial Prediction

Comparative Analysis of Machine Learning Architectures
for Satellite-Based Surface Monitoring

Case Study: Armero, Colombia

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Part I

Phase 1: Problem Definition

1 Introduction

The municipality of Armero, located in the department of Tolima, Colombia, was the site of one of the deadliest volcanic disasters in Latin America. On November 13, 1985, the eruption of the Nevado del Ruiz volcano generated lahars that buried much of the town, causing approximately 25,000 casualties and severe environmental transformation of the surrounding landscape.

This project aims to analyze the current land surface cover of the Armero region using Sentinel-2 multispectral satellite imagery and supervised machine learning techniques. Through spectral analysis and classification models, the study seeks to identify and quantify different land cover classes such as vegetation, croplands, bare soil, and urban areas.

The project follows a GeoAI workflow in which satellite spectral information is transformed into structured datasets for machine learning-based land cover prediction, independent of traditional GIS software.

2 Event Selection

The selected case study corresponds to the Armero region in Tolima, Colombia, affected historically by the Nevado del Ruiz volcanic eruption on November 13, 1985.

2.1 Reasons for Selection

- Historical and environmental relevance: one of the worst volcanic disasters in Latin American history.
- Presence of heterogeneous land cover types: vegetation, croplands, bare soil, and urban zones.
- Availability of cloud-free Sentinel-2 L2A imagery.
- Suitable spectral diversity for supervised classification tasks.

The analysis focuses on identifying present-day surface conditions and understanding the spatial distribution of land cover categories within the study area.

3 Study Area

The Region of Interest (ROI) is centered on the former municipality of Armero, Tolima, Colombia, now known as Armero Guayabal.

3.1 ROI Coordinates

Parameter	**Value**
Minimum Longitude	$-74.925728°$ W
Maximum Longitude	$-74.834747°$ W
Minimum Latitude	$4.997067°$ N
Maximum Latitude	$5.086328°$ N
Approximate Center	$5.042°$ N, $-74.880°$ W
Total Area	$\approx 100.1\ \text{km}^2$ (10 km $\times$ 10 km)

Table 1: ROI coordinates defined as a GeoJSON polygon

3.2 Study Area Characteristics

The selected ROI covers approximately 10 km $\times$ 10 km, tightly centered on the former urban area of Armero and its immediate surroundings. It contains a diverse range of land cover types:

- Agricultural areas and croplands in the flat zones of the Magdalena River valley.
- Natural vegetation and forest patches on surrounding hillsides.
- Bare soil and exposed terrain, including lahar-affected zones still visible after the 1985 disaster.
- Small urban areas corresponding to Armero Guayabal.

This diversity of spectral responses makes the region suitable for supervised machine learning classification experiments.

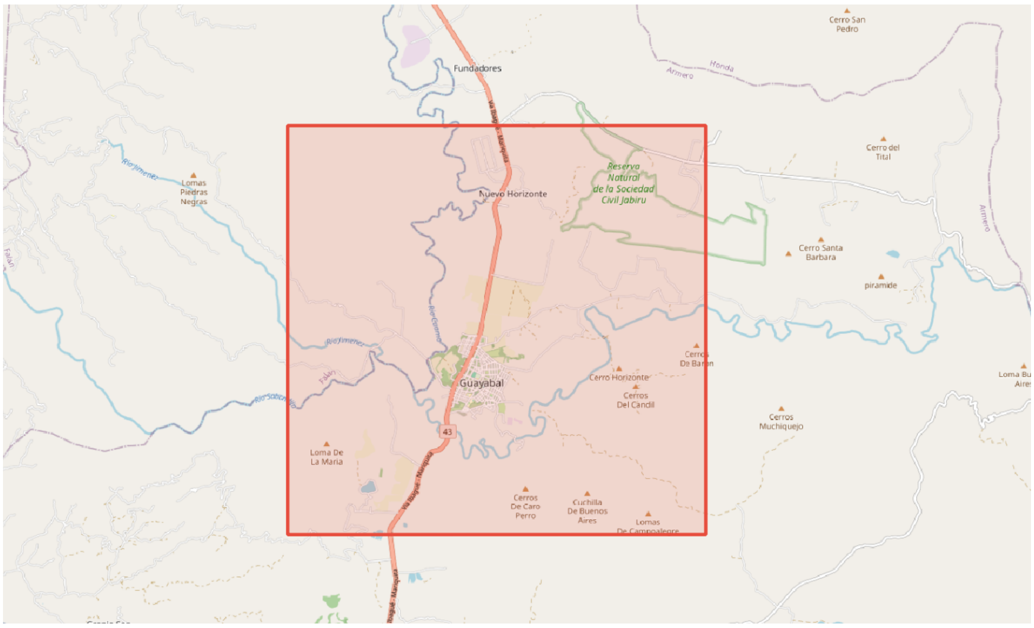

Figure 1: Region of Interest (ROI) centered on Armero Guayabal, Tolima, Colombia. The rectangle covers approximately 100.1 km^2 (10 km × 10 km).

4 Satellite Imagery Metadata

The project uses multispectral imagery from the Sentinel-2 mission developed by the Copernicus Programme (ESA).

4.1 Selected Satellite Product

Attribute	**Value**
Product Name	S2A_MSIL2A_20240815T152651_ N0511_R025_T18NWL
Acquisition Date	August 15, 2024
Sensing Time	15:26:51 UTC
Processing Level	Level-2A (Bottom of Atmosphere)
Tile	T18NWL
Cloud Coverage	15%
Satellite	Sentinel-2A
Sensor	MSI (Multispectral Instrument)
Coordinate System	WGS84 / UTM Zone 18N (EPSG:32618)
Image Format	GeoTIFF (.tif)

Table 2: Sentinel-2 product metadata

Figure 2: Full Sentinel-2 scene (`armero_stack.tif`) covering tile T18NWL before clipping to the ROI. True color composite (B4-B3-B2).

4.2 Spectral Bands Used

Band	Description	Native Resolution
B2	Blue	10 m
B3	Green	10 m
B4	Red	10 m
B5	Vegetation Red Edge 1	20 m (resampled to 10 m)
B6	Vegetation Red Edge 2	20 m (resampled to 10 m)
B7	Vegetation Red Edge 3	20 m (resampled to 10 m)
B8	Near Infrared (NIR)	10 m
B8A	Narrow Near Infrared	20 m (resampled to 10 m)
B11	Short Wave Infrared 1 (SWIR)	20 m (resampled to 10 m)
B12	Short Wave Infrared 2 (SWIR)	20 m (resampled to 10 m)

Table 3: Spectral bands included in the multi-band stack

Bands at 20 m resolution (B5, B6, B7, B8A, B11, B12) were resampled to 10 m using bilinear interpolation via the `resample()` function from the `terra` package in R, ensuring spatial consistency across all bands.

5 Dataset Creation Methodology

The dataset creation process follows a supervised classification workflow based on multi-spectral data extraction from Sentinel-2 L2A imagery acquired over the Armero region.

5.1 Methodology Overview

1. Download of Sentinel-2 Level-2A imagery from the Copernicus Data Space (`https://dataspace.copernicus.eu/`) for the Armero Guayabal region, selecting the August 15, 2024 scene with 15% cloud coverage.

2. Stacking of individual spectral bands (JP2 format) into a single multi-band GeoTIFF using the `terra` package in R (`fase1_stack.R`). Bands at 20 m resolution (B5, B6, B7, B8A, B11, B12) were resampled to 10 m to match the native resolution of B2, B3, B4, and B8, producing the file `armero_stack.tif`.

3. Clipping of the multi-band stack to the ROI extent using a Python script (`fase1_clip.py`) with the `rasterio` library. The ROI was defined as a GeoJSON polygon in WGS84 (EPSG:4326) and reprojected to UTM Zone 18N (EPSG:32618) prior to clipping via `pyproj`, generating the file `armero_recortado.tif`.

4. Digitization of training polygons in QGIS over the clipped image visualized in true color composite (R:B4, G:B3, B:B2), identifying four land cover classes: *vegetacion*, *cultivos*, *suelo_desnudo*, and *urbano*. Polygons were saved as a GeoPackage file (`muestras_armero.gpkg`).

5. Extraction of pixel-level spectral values within training polygons using the `extract()` function from the `terra` package in R (`fase1_dataset.R`), including geographic coordinates (x, y).

6. Balanced random sampling of 2,000 pixels per class (8,000 total) using the `dplyr` package (`slice_sample()`), ensuring class balance and computational efficiency.

7. Export of the final dataset in Tab-Separated Values (TSV) format as `training_data.tsv`.

Figure 3: Clipped Sentinel-2 image (`armero_recortado.tif`) in true color composite (B4-B3-B2). Green areas correspond to dense vegetation; light brown areas correspond to bare soil and croplands; the white cluster near center is Armero Guayabal.

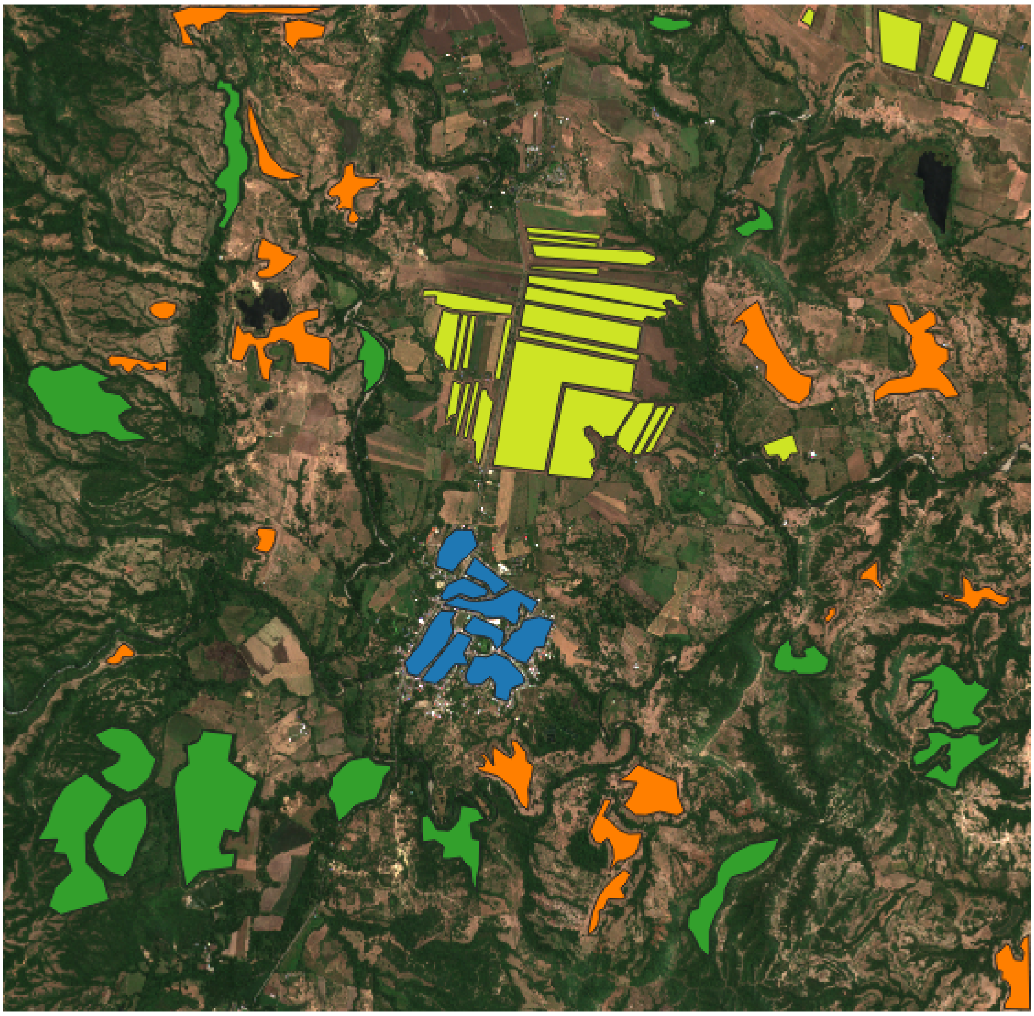

Figure 4: Training polygons digitized over the Armero study area, color-coded by land cover class: *vegetacion* (green), *cultivos* (orange), *suelo_desnudo* (black), and *urbano* (white).

5.2 Land Cover Classes

Class	**Available Pixels**	**Pixels Sampled**	**Percentage**
vegetacion	40,077	2,000	25%
cultivos	34,370	2,000	25%
suelo_desnudo	23,721	2,000	25%
urbano	9,553	2,000	25%
Total	**107,721**	**8,000**	**100%**

Table 4: Available and sampled pixels per land cover class

Note: Water bodies were not included as a distinct class due to the difficulty of reliably delineating narrow river channels at 10 m spatial resolution within this study area.

6 Dataset Structure

The generated TSV file `training_data.tsv` contains 8,000 rows, each representing an individual pixel extracted from the training polygons. The dataset includes the following columns:

Column	Description
x	Pixel easting coordinate in UTM Zone 18N (EPSG:32618)
y	Pixel northing coordinate in UTM Zone 18N (EPSG:32618)
B2	Blue band surface reflectance (10 m)
B3	Green band surface reflectance (10 m)
B4	Red band surface reflectance (10 m)
B5	Red Edge 1 surface reflectance (resampled to 10 m)
B6	Red Edge 2 surface reflectance (resampled to 10 m)
B7	Red Edge 3 surface reflectance (resampled to 10 m)
B8	Near Infrared surface reflectance (10 m)
B8A	Narrow NIR surface reflectance (resampled to 10 m)
B11	SWIR 1 surface reflectance (resampled to 10 m)
B12	SWIR 2 surface reflectance (resampled to 10 m)
clase	Ground truth land cover label

Table 5: Structure of the TSV training dataset

7 Phase 1 Conclusion

This first phase establishes the conceptual and technical foundation of the project by defining the study area, selecting the satellite imagery, and designing the methodology for spectral data extraction. The ROI covers approximately 100.1 km^2 tightly centered on Armero Guayabal, providing a focused and spectrally diverse scene for classification. The generated dataset of 8,000 balanced pixels across four land cover classes provides a robust and representative training set for the subsequent machine learning phases.

Part II

Phase 2: Pre-processing and Training

8 Introduction

This second phase focuses on the implementation and evaluation of five supervised machine learning models for land cover classification over the Armero study area. Each model was trained on the balanced spectral dataset generated in Phase 1, which contains 8,000 pixels (2,000 per class) extracted from Sentinel-2 L2A imagery. The dataset was split into 70% for training (5,600 pixels) and 30% for validation (2,400 pixels).

For each model, a **3 × 3 Hyperparameter Optimization (HPO)** grid was applied using 5-fold cross-validation. This systematic search evaluated all nine combinations of two key hyperparameters (three values each), selecting the configuration with the highest cross-validation accuracy before final evaluation on the held-out test set.

All models were evaluated using: Overall Accuracy, Kappa Coefficient, Precision, Recall, and F1-Score.

9 Data Preparation

9.1 Dataset Split

Subset	**Pixels**	**Percentage**
Training	5,600	70%
Validation	2,400	30%
Total	**8,000**	**100%**

Table 6: Training and validation split (stratified, `set.seed(123)`)

9.2 Data Normalization

Models sensitive to feature scale were trained on a rescaled version of the dataset. The ANN and KNN models used min-max normalization, rescaling each spectral band to the range [0, 1]:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

The SVM was trained on standardized features (zero mean, unit variance), $z = (x - \mu)/\sigma$, as required by its RBF kernel. Decision Trees and Naive Bayes were trained on the original unnormalized values, since these models are invariant to monotonic scaling.

10 Model 1: Decision Tree (DT)

10.1 Description

A Decision Tree partitions the spectral feature space through hierarchical binary splits. Each node selects the band and threshold that maximizes class separation. The complexity parameter `cp` controls pruning (lower values allow deeper trees), while `maxdepth` sets a hard limit on tree depth.

10.2 Hyperparameter Optimization

HPO was performed using 5-fold cross-validation over a 3×3 grid of `cp` $\times$ `maxdepth`:

maxdepth	**cp**	**Accuracy (CV)**	**Kappa (CV)**
5	0.001	90.39%	0.8719
5	0.010	88.59%	0.8479
5	0.100	87.57%	0.8343
10	0.001	92.14%	0.8952
10	0.010	88.59%	0.8479
10	0.100	87.57%	0.8343
20	**0.001**	**92.34%**	**0.8979**
20	0.010	88.59%	0.8479
20	0.100	87.57%	0.8343

Table 7: 3×3 HPO grid results for Decision Tree. Best configuration in bold.

The optimal configuration was **cp = 0.001, maxdepth = 20**. Lower `cp` values consistently outperformed larger ones regardless of depth, indicating that aggressive pruning hurts performance on this spectrally diverse dataset.

10.3 Confusion Matrix

Predicted \ Reference	cultivos	suelo_desnudo	urbano	vegetacion
cultivos	492	45	8	0
suelo_desnudo	104	535	14	0
urbano	4	20	576	0
vegetacion	0	0	2	600

Table 8: Confusion matrix for the optimized Decision Tree (cp = 0.001, maxdepth = 20)

10.4 Performance Metrics

Class	Precision	Recall	F1-Score
cultivos	0.9028	0.8200	0.8594
suelo_desnudo	0.8193	0.8917	0.8540
urbano	0.9600	0.9600	0.9600
vegetacion	0.9967	1.0000	0.9983

Table 9: Per-class performance metrics for the optimized Decision Tree

Metric	Value
Overall Accuracy	91.79%
Kappa Coefficient	0.8906

Table 10: Overall metrics for the optimized Decision Tree

11 Model 2: Support Vector Machine (SVM)

11.1 Description

A Support Vector Machine is a margin-based classifier that searches for the hyperplane maximizing the separation between classes in the spectral feature space. A Radial Basis Function (RBF) kernel was used to handle the non-linear separability of land cover classes. The cost parameter `C` controls the trade-off between margin width and training misclassifications, while `sigma` governs the width of the RBF kernel. All bands were standardized prior to training.

11.2 Hyperparameter Optimization

HPO was performed using 5-fold cross-validation over a 3 × 3 grid of `C` × `sigma`:

C	sigma	Accuracy (CV)	Kappa (CV)
0.1	0.01	89.02%	0.8536
1.0	0.01	91.46%	0.8862
10.0	0.01	92.89%	0.9052
0.1	0.10	91.29%	0.8838
1.0	0.10	94.05%	0.9207
10.0	0.10	95.77%	0.9436
0.1	1.00	92.62%	0.9017
1.0	1.00	95.75%	0.9433
10.0	**1.00**	**96.91%**	**0.9588**

Table 11: 3 × 3 HPO grid results for SVM (RBF kernel). Best configuration in bold.

The optimal configuration was **C = 10, sigma = 1.0**. Accuracy increased monotonically with both `C` and `sigma`, confirming that the spectral classes benefit from a wide-kernel, high-margin model.

11.3 Confusion Matrix

Predicted \ Reference	cultivos	suelo_desnudo	urbano	vegetacion
cultivos	545	9	0	0
suelo_desnudo	55	584	6	0
urbano	0	7	593	0
vegetacion	0	0	1	600

Table 12: Confusion matrix for the optimized SVM (C = 10, sigma = 1.0)

11.4 Performance Metrics

Class	Precision	Recall	F1-Score
cultivos	0.9838	0.9083	0.9445
suelo_desnudo	0.9054	0.9733	0.9382
urbano	0.9883	0.9883	0.9883
vegetacion	0.9983	1.0000	0.9992

Table 13: Per-class performance metrics for the optimized SVM

Metric	Value
Overall Accuracy	96.75%
Kappa Coefficient	0.9567

Table 14: Overall metrics for the optimized SVM

12 Model 3: Artificial Neural Network (ANN)

12.1 Description

An Artificial Neural Network (ANN) is a computational model composed of interconnected layers of neurons that learn non-linear relationships between spectral signatures and land cover categories. The architecture used is a single hidden layer feedforward network implemented via the `nnet` package in R. The hyperparameter `size` controls the number of neurons in the hidden layer, while `decay` is an L2 regularization term that prevents overfitting. Input bands were normalized to [0, 1] prior to training.

12.2 Hyperparameter Optimization

HPO was performed using 5-fold cross-validation over a 3×3 grid of `size` $\times$ `decay`:

size	decay	Accuracy (CV)	Kappa (CV)
5	0.001	96.23%	0.9498
5	0.010	96.32%	0.9510
5	0.100	94.52%	0.9269
10	0.001	97.04%	0.9605
10	0.010	97.23%	0.9631
10	0.100	95.04%	0.9338
20	0.001	97.32%	0.9643
20	**0.010**	**97.46%**	**0.9662**
20	0.100	94.98%	0.9331

Table 15: 3×3 HPO grid results for ANN. Best configuration in bold.

The optimal configuration was **size = 20, decay = 0.01**. Larger networks (size = 20) consistently outperformed smaller ones, and moderate regularization (decay = 0.01) proved superior to both very low and very high values.

12.3 Confusion Matrix

Predicted \ Reference	cultivos	suelo_desnudo	urbano	vegetacion
cultivos	555	16	0	0
suelo_desnudo	45	578	3	0
urbano	0	6	597	0
vegetacion	0	0	0	600

Table 16: Confusion matrix for the optimized ANN (size = 20, decay = 0.01)

12.4 Performance Metrics

Class	Precision	Recall	F1-Score
cultivos	0.9720	0.9250	0.9479
suelo_desnudo	0.9233	0.9633	0.9429
urbano	0.9900	0.9950	0.9925
vegetacion	1.0000	1.0000	1.0000

Table 17: Per-class performance metrics for the optimized ANN

Metric	Value
Overall Accuracy	97.08%
Kappa Coefficient	0.9611

Table 18: Overall metrics for the optimized ANN

13 Model 4: K-Nearest Neighbor (KNN)

13.1 Description

K-Nearest Neighbor (KNN) is an instance-based, non-parametric classifier that assigns the label of a new pixel according to the majority class among its k nearest neighbors in the spectral feature space. The `kknn` implementation was used, which supports both the number of neighbors `k` and the distance metric as tunable parameters. All input bands were normalized to [0, 1] before training, as distance-based methods are highly sensitive to feature scale.

13.2 Hyperparameter Optimization

HPO was performed using 5-fold cross-validation over a 3×3 grid of `k` $\times$ distance metric:

k	Distance	Accuracy (CV)	Kappa (CV)
3	Manhattan	94.77%	0.9302
3	Euclidean	95.14%	0.9352
3	Minkowski	95.05%	0.9340
7	Manhattan	94.77%	0.9302
7	Euclidean	95.14%	0.9352
7	Minkowski	95.05%	0.9340
15	Manhattan	94.77%	0.9302
15	**Euclidean**	**95.14%**	**0.9352**
15	Minkowski	95.05%	0.9340

Table 19: 3×3 HPO grid results for KNN. Best configuration in bold (tie broken by larger k).

The optimal configuration was **k = 15, Euclidean distance**. Notably, the value of `k` had no influence on accuracy — only the distance metric mattered, with Euclidean consistently outperforming Manhattan and Minkowski. This suggests that with well-normalized spectral data and clearly distinct classes, the choice of neighborhood size is irrelevant.

13.3 Confusion Matrix

Predicted \ Reference	cultivos	suelo_desnudo	urbano	vegetacion
cultivos	555	40	5	0
suelo_desnudo	44	549	14	0
urbano	1	11	577	1
vegetacion	0	0	4	599

Table 20: Confusion matrix for the optimized KNN (k = 15, Euclidean distance)

13.4 Performance Metrics

Class	Precision	Recall	F1-Score
cultivos	0.9250	0.9250	0.9250
suelo_desnudo	0.9044	0.9150	0.9097
urbano	0.9780	0.9617	0.9697
vegetacion	0.9934	0.9983	0.9958

Table 21: Per-class performance metrics for the optimized KNN

Metric	Value
Overall Accuracy	95.00%
Kappa Coefficient	0.9333

Table 22: Overall metrics for the optimized KNN

14 Model 5: Naive Bayes (NB)

14.1 Description

Naive Bayes is a probabilistic classifier based on Bayes' Theorem that assumes conditional independence between spectral bands given the class. The kernel density estimation variant (`usekernel = TRUE`) was used throughout, which models each band's distribution non-parametrically and is better suited for the non-Gaussian spectral distributions typical of satellite imagery. The hyperparameter `laplace` controls additive smoothing, and `adjust` sets the bandwidth multiplier for the kernel density estimator.

14.2 Hyperparameter Optimization

HPO was performed using 5-fold cross-validation over a 3×3 grid of `laplace` × `adjust`:

laplace	adjust	Accuracy (CV)	Kappa (CV)
0.0	0.5	87.04%	0.8271
0.0	**1.0**	**87.29%**	**0.8305**
0.0	2.0	87.02%	0.8269
0.5	0.5	87.04%	0.8271
0.5	1.0	87.29%	0.8305
0.5	2.0	87.02%	0.8269
1.0	0.5	87.04%	0.8271
1.0	1.0	87.29%	0.8305
1.0	2.0	87.02%	0.8269

Table 23: 3 × 3 HPO grid results for Naive Bayes. Best configuration in bold (tie broken by laplace = 0).

The optimal configuration was **laplace = 0, adjust = 1.0**. The `laplace` parameter had no effect on performance: with 2,000 samples per class, zero-frequency problems do not arise, making smoothing redundant. Only `adjust` produced marginal differences.

14.3 Confusion Matrix

Predicted \ Reference	cultivos	suelo_desnudo	urbano	vegetacion
cultivos	422	18	12	0
suelo_desnudo	138	543	50	0
urbano	40	39	530	3
vegetacion	0	0	8	597

Table 24: Confusion matrix for the optimized Naive Bayes (laplace = 0, adjust = 1.0)

14.4 Performance Metrics

Class	Precision	Recall	F1-Score
cultivos	0.9336	0.7033	0.8023
suelo_desnudo	0.7428	0.9050	0.8159
urbano	0.8660	0.8833	0.8746
vegetacion	0.9868	0.9950	0.9909

Table 25: Per-class performance metrics for the optimized Naive Bayes

Metric	Value
Overall Accuracy	87.17%
Kappa Coefficient	0.8289

Table 26: Overall metrics for the optimized Naive Bayes

15 Comparative Summary

Model	Best Parameters	Accuracy	Kappa	F1 mean	Weakest class
ANN	size=20, decay=0.01	**97.38%**	**0.9650**	**0.9737**	suelo_desnudo (0.9429)
SVM	C=10, sigma=1.0	96.75%	0.9567	0.9676	suelo_desnudo (0.9382)
KNN	k=15, Euclidean	95.00%	0.9333	0.9501	suelo_desnudo (0.9097)
Decision Tree	cp=0.001, maxdepth=20	91.79%	0.8906	0.9179	suelo_desnudo (0.8540)
Naive Bayes	laplace=0, adjust=1.0	87.17%	0.8289	0.8709	cultivos (0.8023)

Table 27: Comparative performance summary across the five classification models. Accuracy values from `fase2_resumen.R`; F1 mean is the unweighted average across the four land cover classes.

16 Phase 2 Conclusion

Among the five models evaluated, the Artificial Neural Network achieved the best overall performance with 97.38% accuracy and a Kappa coefficient of 0.9650, followed closely by the Support Vector Machine (96.75%, Kappa 0.9567). Both non-linear models substantially outperformed the simpler classifiers, confirming that the spectral classes of the Armero study area are not linearly separable and require flexible decision boundaries.

The K-Nearest Neighbor classifier ranked third (95.00%, Kappa 0.9333), while the Decision Tree obtained 91.79% (Kappa 0.8906). Naive Bayes performed the weakest (87.17%, Kappa 0.8289), a result consistent with its conditional-independence assumption, which does not hold for the highly correlated Sentinel-2 spectral bands.

Across all five models, *vegetacion* was the most reliably classified category (F1 $\geq$ 0.98 in every model), while *suelo_desnudo* and *cultivos* consistently showed the strongest spectral overlap and accounted for most misclassifications. This is expected, as both surfaces share similar visible and short-wave infrared reflectance values within the study area.

The ANN (size = 20, decay = 0.01) is confirmed as the best-performing model and will be used for full-scene thematic inference in Phase 3.

Part III

Phase 3: Synthesis and Prediction

17 Introduction

This final phase synthesizes the results of the comparative analysis carried out in Phase 2 and deploys the selected classification model to produce a complete thematic land-cover map of the Armero study area. Unlike the training phase, which operated on a discrete set of 8,000 pixels extracted from training polygons, the inference phase applies the trained model to the entire multispectral raster stack, classifying every valid pixel within the Region of Interest.

The objectives of this phase are threefold: (i) to confirm the selection of the most robust classification architecture based on the validation metrics obtained in Phase 2; (ii) to perform full-scene inference and export the result as a georeferenced GeoTIFF; and (iii) to generate a thematic visualization and quantify the surface area occupied by each land cover class.

18 Model Selection for Inference

18.1 Selection Criteria

To ensure a rigorous and defensible choice, the final model was selected following an empirical, multi-criteria procedure rather than relying on a single number. The criteria were applied in the following order of priority:

1. **Macro F1-Score.** Prioritized over Overall Accuracy because it balances precision and recall across all classes and is more informative when spectrally similar classes

(such as *cultivos* and *suelo_desnudo*) are present.

2. **Kappa Coefficient.** As a secondary criterion, it measures the agreement between predictions and reference data beyond what would be expected by chance, providing a more conservative estimate of model quality than raw accuracy.

3. **Per-class consistency.** When two models obtained comparable F1 and Kappa values, the confusion matrices were inspected class by class, favouring the model with the most balanced performance across all four classes.

4. **Robustness over peak performance.** A model with consistent performance across all classes was preferred over one with a higher global score but a weak class, since spatial consistency in the final map is more valuable than a marginal gain in a single aggregate metric.

18.2 Selected Model

Applying these criteria, the **Artificial Neural Network (ANN)** was selected for full-scene inference. It obtained the highest values in every aggregate metric simultaneously: the best Macro F1-Score (0.9737), the highest Overall Accuracy (97.38%), and the highest Kappa coefficient (0.9650). Crucially, it was also the most balanced model at the per-class level, with no class falling below an F1-Score of 0.94 and *vegetacion* reaching a perfect F1 of 1.0000. The Support Vector Machine was a close competitor, but the ANN matched or exceeded it on every criterion, making the choice unambiguous in this case.

Attribute	**Value**
Selected model	Artificial Neural Network (ANN)
Optimal hyperparameters	size = 20, decay = 0.01
Final Overall Accuracy	97.38%
Final Kappa Coefficient	0.9650
Macro F1-Score	0.9737

Table 28: Summary of the model selected for full-scene inference

19 Full-Scene Inference

19.1 Inference Workflow

The inference procedure was implemented in R (`inferencia.R`) using the `terra` and `caret` packages. Since the trained model object was not serialized during Phase 2, the ANN was re-trained at the start of the inference script using the same random seed (`set.seed(123)`), the same 70% training partition, and the same min-max normalization parameters derived from that partition. This guarantees that the reflectance values of the full raster are scaled identically to the data the model was trained on.

The clipped multispectral raster (`armero_recortado.tif`, 987×1009 pixels) was loaded, its ten bands (B2–B12) were normalized using the training minima and maxima, and the model assigned a class label to each valid pixel. Pixels containing no-data values in any band were excluded from prediction and preserved as no-data in the output. A total of **995,883** valid pixels were classified.

19.2 Output Raster Specifications

The predicted map was exported as a georeferenced GeoTIFF (`armero_predicho.tif`) that preserves the spatial reference of the source imagery:

- **Spatial Consistency:** the output retains the original Coordinate Reference System (WGS84 / UTM Zone 18N, EPSG:32618) and the affine transformation parameters of `armero_recortado.tif`, so that the prediction aligns exactly with the Sentinel-2 imagery.
- **Data type:** single-band unsigned 8-bit integer raster (`INT1U`), where each pixel value corresponds to a land cover class code.
- **Resolution:** 10 m, matching the input stack.

The class-to-code mapping used in the output raster is defined as follows:

Pixel Code	Land Cover Class
1	cultivos (Croplands)
2	suelo_desnudo (Bare Soil)
3	urbano (Urban Area)
4	vegetacion (Vegetation)

Table 29: Integer codes assigned to each land cover class in the predicted raster

20 Thematic Map

20.1 Color Symbology

The predicted GeoTIFF was imported into QGIS and rendered using a paletted/unique-values symbology so that each class is clearly distinguishable. The color scheme applied is:

Code	Class	Color
1	cultivos	Light green
2	suelo_desnudo	Orange
3	urbano	Blue
4	vegetacion	Green

Table 30: Discrete color symbology applied to the thematic land-cover map

20.2 Resulting Map

Figure 5 shows the final classified map over an OpenStreetMap base layer. Vegetation (green) dominates the hillsides surrounding the valley, while bare soil (orange) and croplands (light green) concentrate in the flatter central zone along the Magdalena River. The urban class (blue) appears mainly in the central-south sector, corresponding to the present-day settlement of Armero Guayabal and the area historically buried by the 1985 lahars.

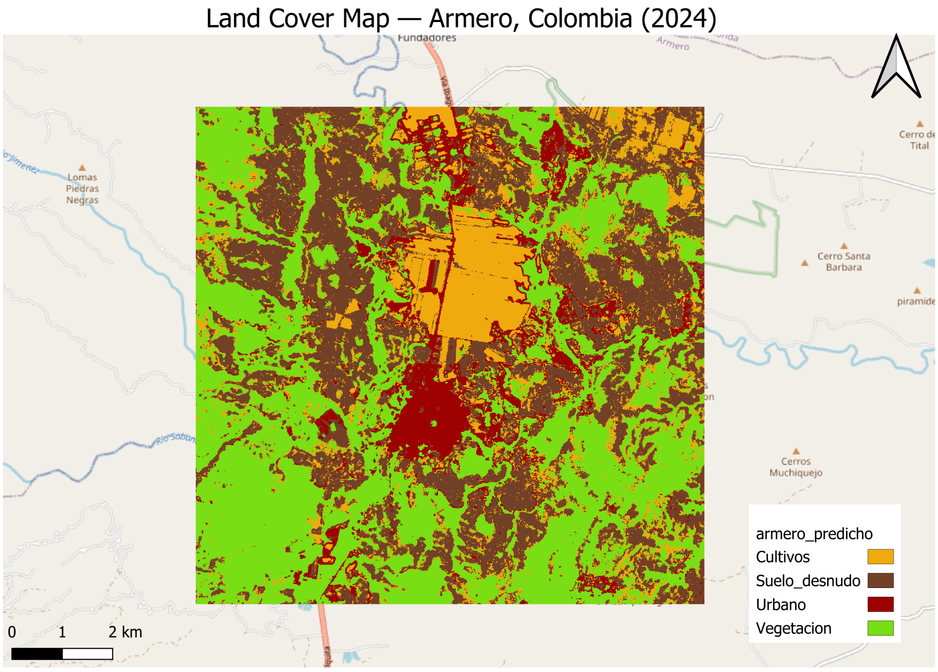

Figure 5: Predicted land-cover map of the Armero study area, obtained by applying the optimized ANN model to the full Sentinel-2 scene. Green: vegetation; light green: croplands; orange: bare soil; blue: urban areas.

21 Area Quantification

The total surface area occupied by each class was computed from the predicted raster by counting the number of pixels per class and multiplying by the pixel area. At a 10 m resolution, each pixel covers $10 \times 10 = 100$ m^2 $= 0.01$ hectares, so the area per class is obtained as:

$$A_{\text{class}} = N_{\text{pixels}} \times 0.01 \text{ ha} \tag{2}$$

Class	**Pixel Count**	**Area (ha)**	**Area (km^2)**	**Area (%)**
cultivos	167,810	1,678.10	16.78	16.85
suelo_desnudo	316,685	3,166.85	31.67	31.80
urbano	101,474	1,014.74	10.15	10.19
vegetacion	409,914	4,099.14	40.99	41.16
Total	**995,883**	**9,958.83**	**99.59**	**100.00**

Table 31: Surface area occupied by each land cover class in the Armero study area

Vegetation is the most extensive class, covering 40.99 km^2 (41.16% of the classified area), followed by bare soil with 31.67 km^2 (31.80%). Croplands account for 16.78 km^2 (16.85%) and the urban class for 10.15 km^2 (10.19%).

22 Phase 3 Conclusion

This phase completed the full remote-sensing classification pipeline, from the selection of the optimal model to the generation of a quantitative thematic product. The Artificial Neural Network, selected empirically as the best-performing architecture in Phase 2 (97.38% accuracy, Kappa 0.9650, Macro F1 0.9737), was deployed over the entire Sentinel-2 scene to classify 995,883 pixels into four land cover categories.

The resulting map indicates that the Armero study area is currently dominated by natural vegetation (41.16%), distributed mainly across the surrounding hillsides, followed by bare soil (31.80%) and croplands (16.85%) concentrated in the flat central zone of the Magdalena River valley. The urban class (10.19%) is largely confined to the central-south sector around Armero Guayabal. Nearly four decades after the 1985 disaster, the landscape thus appears largely recovered in vegetative terms, while the central plain still exhibits a substantial proportion of exposed soil and agricultural use.

A noteworthy observation is the spatial dispersion of the *suelo_desnudo* class, which appears scattered throughout the scene rather than concentrated in a single region. This pattern suggests that the model may be conflating bare soil with spectrally similar surfaces such as unpaved roads, harvested or fallow croplands, and transitional terrain. This behaviour is consistent with the validation results of Phase 2, where *suelo_desnudo* obtained the lowest per-class F1-Score (0.9429) and showed the strongest confusion with *cultivos*, owing to their overlapping reflectance in the visible and short-wave infrared bands. While this residual confusion is the main source of uncertainty in the final product, it does not compromise the overall quality of the classification, which remains high across all aggregate metrics.

Overall, this project demonstrates a complete GeoAI workflow, transforming raw multispectral reflectance into an actionable, georeferenced land-cover map and quantitative area statistics, and illustrates both the strengths and the inherent spectral limitations of supervised classification applied to a historically significant region.